

Quantum Machine Learning

QNN, QSVM, HHL, QAOA, VQE

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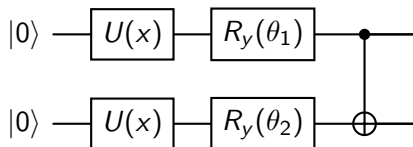
Spring 2026

$$|\psi(x, \theta)\rangle = U(\theta)U(x)|0\rangle$$

$$f(x, \theta) = \langle\psi(x, \theta)|O|\psi(x, \theta)\rangle$$

- $U(x)$: feature map
- $U(\theta)$: variational ansatz
- O : observable

Circuit Representation



Pauli Expansion of the Ansatz

$$U(\theta) = e^{-iH(\theta)}$$

$$H(\theta) = \sum_{\alpha} \theta_{\alpha} P_{\alpha}$$

- $P_{\alpha} \in \{I, X, Y, Z\}^{\otimes n}$
- Equivalent to many-body Hamiltonian

Feature Map as Hamiltonian Evolution

$$U(x) = e^{-iH(x)}, \quad H(x) = \sum_i x_i Z_i + \sum_{ij} x_i x_j Z_i Z_j$$

- Data = coupling constants
- Embedding = time evolution

$$U^\dagger O U = O + i[H, O] + \frac{i^2}{2}[H, [H, O]] + \dots$$

- Generates hierarchical correlations
- Analogous to coupled-cluster expansions

Effective Operator Structure

$$O_{\text{eff}} = U^\dagger O U$$

- Encodes nonlinear transformations
- Depth $\rightarrow$ higher-order commutators

$$\partial_{\theta} f = \langle \psi | i[H_{\theta}, O] | \psi \rangle$$

- Gradients = commutators
- Physical interpretation: response functions

Parameter Shift Rule

$$\frac{\partial f}{\partial \theta} = \frac{1}{2} \left[f\left(\theta + \frac{\pi}{2}\right) - f\left(\theta - \frac{\pi}{2}\right) \right]$$

- Exact gradient
- Measurement-based evaluation

$$F_{ij} = 4 \operatorname{Re} (\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle)$$

$$ds^2 = \sum_{ij} F_{ij} d\theta_i d\theta_j$$

- Defines Riemannian geometry
- Distance between quantum states

$$\dot{\theta} = F^{-1} \nabla C$$

- Corrects Euclidean gradient
- Respects quantum geometry

Time-Dependent Variational Principle

$$\delta \|(i\partial_t - H)|\psi\rangle\| = 0$$

Project Schrödinger equation onto tangent space:

$$\langle \partial_i \psi | (i\partial_t - H) | \psi \rangle = 0$$

$$\sum_j F_{ij} \dot{\theta}_j = C_i$$

$$C_i = \text{Im} \langle \partial_i \psi | H | \psi \rangle$$

Interpretation of TDVP

- QNN training = classical dynamical system
- Parameters evolve like generalized coordinates

$$C(\theta) = \sum_i (f(x_i, \theta) - y_i)^2$$

- Highly non-convex
- Structured by geometry

$$\text{Var}(\nabla C) \sim \frac{1}{2^n}$$

- Gradient vanishes exponentially

Origin of Barren Plateaus

- Random circuits $\rightarrow$ unitary 2-designs
- Expectation values concentrate

Avoiding Plateaus

- Local cost functions
- Problem-inspired ansätze
- Layerwise training

$$S(\rho_A) = -\text{Tr}(\rho_A \log \rho_A)$$

- Controls complexity

Expressivity Regimes

- Low entanglement $\rightarrow$ classical-like
- Moderate $\rightarrow$ optimal learning
- High $\rightarrow$ concentration of measure

Quantum Neural Tangent Kernel

$$K(x, x') = \sum_i \frac{\partial f(x)}{\partial \theta_i} \frac{\partial f(x')}{\partial \theta_i}$$

Interpretation of NTK

- Linearized training regime
- Connects QNN $\leftrightarrow$ kernel methods

$$U(\theta) = \mathcal{T}e^{-i \int H(t)dt}$$

- θ = control parameters
- Learning = control optimization

Summary of QNN Section

- QNN = variational many-body state
- Geometry defined by QFI
- Training = TDVP dynamics
- Expressivity tied to entanglement

Variational Principle

Given Hamiltonian H :

$$E_0 = \min_{\psi} \langle \psi | H | \psi \rangle$$

For trial state $|\psi(\theta)\rangle$:

$$E(\theta) \geq E_0$$

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

$$U(\theta) = e^{-iH(\theta)}, \quad H(\theta) = \sum_{\alpha} \theta_{\alpha} P_{\alpha}$$

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

- Optimization in parameter space

Hamiltonian Decomposition

$$H = \sum_i c_i P_i$$

- P_i : Pauli strings
- Measurement reduces to expectation values

$$E(\theta) = \sum_i c_i \langle P_i \rangle$$

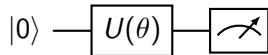
- Requires repeated measurements

$$\frac{\partial E}{\partial \theta} = i \langle \psi | [H_\theta, H] | \psi \rangle$$

- Structure similar to QNN gradients

Parameter Shift Rule

$$\frac{\partial E}{\partial \theta} = \frac{1}{2} \left[E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) \right]$$



$$\sum_j F_{ij} \dot{\theta}_j = C_i$$

- VQE optimization = projected dynamics

Unitary Coupled Cluster (UCC)

$$U = e^{T - T^\dagger}$$

$$T = \sum_{ai} t_{ai} a_a^\dagger a_i + \dots$$

- Systematic ansatz

$$e^{A+B} \approx e^A e^B$$

- Used to implement UCC

$$e^{-T} H e^T = H + [H, T] + \frac{1}{2} [[H, T], T] + \dots$$

- Connection to coupled cluster theory

- Ansatz determines reachable states
- Tradeoff between depth and trainability

Noise and Sampling

- Measurement noise affects gradients
- Shot complexity important

Summary

- VQE = variational ground-state solver
- Connected to QNN and TDVP
- Strong link to many-body theory

Optimization Problem

Given cost function $C(z)$:

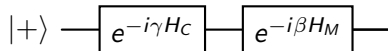
$$H_C|z\rangle = C(z)|z\rangle$$

$$|\gamma, \beta\rangle = \prod_{l=1}^p e^{-i\beta_l H_M} e^{-i\gamma_l H_C} |+\rangle$$

Mixer Hamiltonian

$$H_M = \sum_i X_i$$

Circuit (p=1)



Expectation Value

$$C(\gamma, \beta) = \langle \gamma, \beta | H_C | \gamma, \beta \rangle$$

Connection to Adiabatic Evolution

$$H(s) = (1 - s)H_M + sH_C$$

- QAOA = discretized adiabatic evolution

Trotter Expansion

$$e^{-i(H_M+H_C)t} \approx e^{-iH_M\Delta t}e^{-iH_C\Delta t}$$

$$H_C = \sum_{\langle ij \rangle} \frac{1 - Z_i Z_j}{2}$$

$$\partial_\gamma C = i\langle [H_C, H_{\text{eff}}] \rangle$$

Landscape

- Parameter space (γ, β)
- Non-convex structure

Relation to VQE

- QAOA = structured ansatz
- VQE = general ansatz

- QAOA is a QNN with specific Hamiltonians

Expressivity vs Depth

- Larger $p \rightarrow$ better approximation
- but harder optimization

Summary

- QAOA = variational optimization algorithm
- derived from adiabatic evolution
- connects optimization and quantum dynamics

SVM: Primal Problem

Given data (x_i, y_i) with $y_i \in \{-1, 1\}$:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w \cdot x_i + b) \geq 1$$

Dual Formulation

Introduce Lagrange multipliers α_i :

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{ij} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

Kernel Trick

Replace inner product:

$$x_i \cdot x_j \rightarrow K(x_i, x_j)$$

Implicit mapping:

$$x \rightarrow \phi(x)$$

$$K(x, x') = \langle \phi(x), \phi(x') \rangle$$

If $K(x, x')$ is positive definite:

$$K(x, x') = \sum_k \lambda_k \psi_k(x) \psi_k(x')$$

- Defines embedding into RKHS

Quantum Feature Map

Define:

$$|\phi(x)\rangle = U(x)|0\rangle$$

Then:

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$$

Pauli Expansion of Feature Map

$$U(x) = e^{iH(x)}, \quad H(x) = \sum_{\alpha} f_{\alpha}(x) P_{\alpha}$$

Example:

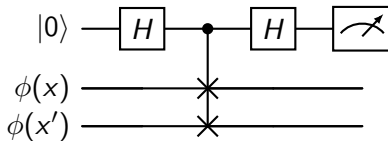
$$H(x) = \sum_i x_i Z_i + \sum_{ij} x_i x_j Z_i Z_j$$

- Equivalent to spin Hamiltonian

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$$

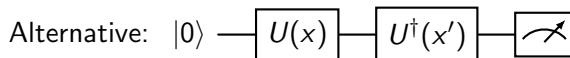
- Transition probability
- Quantum overlap

Swap Test Circuit



$$P(0) = \frac{1 + |\langle \phi(x) | \phi(x') \rangle|^2}{2}$$

Direct Fidelity Circuit



$$P(0) = |\langle \phi(x) | \phi(x') \rangle|^2$$

$$K_{ij} = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$$

- Evaluated on quantum device
- Used in classical optimization

Decision Function

$$f(x) = \sum_i \alpha_i y_i K(x_i, x) + b$$

Expressivity of Quantum Kernels

Kernel determined by:

$$U^\dagger(x)U(x')$$

- Encodes nonlinear correlations
- Depends on circuit depth and entanglement

$$K(x, x') = |\langle 0 | U^\dagger(x) U(x') | 0 \rangle|^2$$

- Equivalent to expectation value

Expressivity vs Entanglement

- No entanglement $\rightarrow$ separable kernel
- Moderate entanglement $\rightarrow$ rich structure
- High entanglement $\rightarrow$ concentration of measure

$$|\langle \phi(x) | \phi(x') \rangle|^2 \approx \frac{1}{2^n}$$

- Kernel becomes nearly constant
- Loss of discriminative power

Quantum Neural Tangent Kernel Connection

$$K_{\text{NTK}}(x, x') = \sum_i \frac{\partial f(x)}{\partial \theta_i} \frac{\partial f(x')}{\partial \theta_i}$$

- QNN induces kernel
- QSVM = explicit kernel method

Kernel methods require solving:

$$(K + \lambda I)\alpha = y$$

- HHL can solve this system
- Leads to fully quantum pipeline

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$$

- Overlap of quantum states
- Analogous to correlation functions

- Data embedded in $\mathcal{H}$
- Kernel defines geometry
- Classification = geometric separation

Advantages

- Access to large Hilbert space
- Non-classical correlations
- Potential quantum advantage

Limitations

- Kernel estimation cost
- Noise sensitivity
- unclear advantage for generic data

Summary

- QSVM = kernel method in Hilbert space
- Kernel = quantum state overlap
- Expressivity tied to entanglement
- Connects QNN and HHL

Problem Statement

We wish to solve the linear system:

$$Ax = b$$

with:

- $A \in \mathbb{C}^{N \times N}$ Hermitian
- $|b\rangle$ efficiently preparable

Goal:

$$|x\rangle \propto A^{-1}|b\rangle$$

Spectral Decomposition

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j|$$

Expand input:

$$|b\rangle = \sum_j \beta_j |u_j\rangle$$

Then:

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

Key Idea of HHL

HHL implements:

$$\lambda_j \longrightarrow \frac{1}{\lambda_j}$$

on the eigenvalues of A .

- This is a **spectral transformation**
- Achieved via Quantum Phase Estimation (QPE)

Quantum Phase Estimation (QPE)

QPE extracts eigenvalues of A :

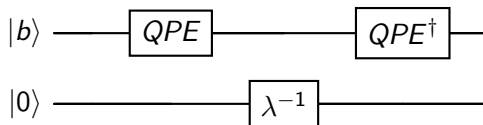
$$U = e^{iAt}$$

$$U|u_j\rangle = e^{i\lambda_j t}|u_j\rangle$$

QPE maps:

$$|u_j\rangle|0\rangle \rightarrow |u_j\rangle|\lambda_j\rangle$$

HHL Circuit Structure



- QPE encodes eigenvalues
- Controlled rotation applies inverse

State After QPE

After QPE:

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle$$

Eigenvalues are stored in a register.

Introduce ancilla qubit:

$$|\lambda_j\rangle|0\rangle \rightarrow |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}}|0\rangle + \frac{C}{\lambda_j}|1\rangle \right)$$

- Encodes λ_j^{-1}

Condition on ancilla $|1\rangle$:

$$|\psi\rangle \propto \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

This is the desired solution state.

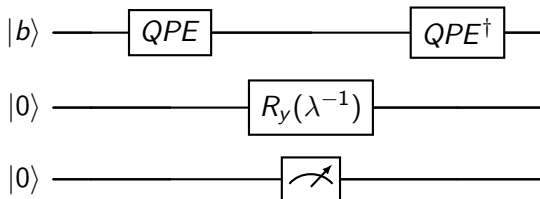
Apply inverse QPE:

$$|\lambda_j\rangle \rightarrow |0\rangle$$

Final state:

$$|x\rangle \propto A^{-1}|b\rangle$$

Full Circuit (Conceptual)



Runtime depends on:

$$\mathcal{O}(\kappa^2 \log N)$$

where:

- κ = condition number
- N = system size

Potential exponential speedup if:

- sparse matrix
- efficient state preparation

Limitations

- Output is quantum state, not full vector
- Requires efficient block encoding
- Sensitive to condition number

$$A^{-1} \sim (\omega I - H)^{-1}$$

This is the **resolvent operator**.

- Appears in Green's functions
- Encodes system response

Connection to Many-Body Physics

$$G(\omega) = (\omega I - H)^{-1}$$

- Poles correspond to eigenvalues
- Residues = spectral weights

HHL performs:

Input state $\rightarrow$ spectral filtering $\rightarrow$ output state

- Filters eigencomponents by λ^{-1}

HHL applies to:

- Linear regression

$$(X^T X)w = X^T y$$

- Kernel methods

$$(K + \lambda I)\alpha = y$$

- Gaussian processes

HHL $\leftrightarrow$ inverse operator

QNN $\leftrightarrow$ variational dynamics

QSVM $\leftrightarrow$ kernel geometry

Summary

- HHL solves $Ax = b$ via spectral decomposition
- Uses QPE + controlled rotations
- Implements inverse operator
- Connects directly to Green's functions

We have studied:

- Variational circuits (QNN, VQE, QAOA)
- Kernel methods (QSVM)
- Linear solvers (HHL)
- PDE/operator learning

Question:

Is there a unifying mathematical structure?

Three Fundamental Structures

Quantum machine learning is built on:

(1) Geometry (2) Dynamics (3) Operators

- Geometry: Hilbert space, QFI
- Dynamics: variational evolution
- Operators: inversion, kernels

Data is embedded as:

$$x \rightarrow |\phi(x)\rangle \in \mathcal{H}$$

Kernel:

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$$

- QSVM = geometry-based learning

QNN:

$$|\psi(\theta)\rangle = e^{-iH(\theta)}|\psi(x)\rangle$$

Training:

$$\sum_j F_{ij} \dot{\theta}_j = C_i$$

- TDVP = projected Schrödinger dynamics

HHL implements:

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle \langle u_j|$$

- Inverse operator
- Spectral filtering

PDE:

$$\mathcal{L}u = f$$

Solution:

$$u = \mathcal{L}^{-1}f$$

$$u(x) = \int G(x, x') f(x') dx'$$

- $G(x, x') =$ Green's function

Transformers as Operator Learners

Transformer:

$$u(x) = \sum_{x'} \alpha(x, x') v(x')$$

- $\alpha(x, x') \approx$ learned kernel
- Approximates Green's function

HHL $\leftrightarrow$ Exact inverse operator

Transformers $\leftrightarrow$ Learned operator

QSVM $\leftrightarrow$ Kernel geometry

QNN $\leftrightarrow$ Variational dynamics

All methods rely on spectral structure:

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j|$$

- HHL: transforms eigenvalues
- QNN: learns eigenstructure
- QSVM: measures overlaps

Learning can be viewed as:

Projection onto submanifold of Hilbert space

- QNN: variational manifold
- TDVP: optimal projection

$$ds^2 = \sum_{ij} F_{ij} d\theta_i d\theta_j$$

- QFI defines curvature
- Learning follows geodesics

Quantum Machine Learning = Geometry + Dynamics + Operators

= Hilbert Space + Variational Evolution + Inverse Maps

Physical Interpretation

- QNN: quantum dynamics under learned Hamiltonian
- QSVM: geometry of quantum states
- HHL: response function / Green's operator
- Transformers: learned propagators

Learning $\equiv$ Understanding and approximating operators

- Exact $\rightarrow$ HHL
- Variational $\rightarrow$ QNN/VQE/QAOA
- Geometric $\rightarrow$ QSVM
- Learned $\rightarrow$ Transformers

We have unified quantum machine learning as:

$$\text{QML} = \text{Geometry} + \text{Dynamics} + \text{Operators}$$

Key question:

Where can genuine quantum advantage emerge?

What is Quantum Advantage?

Quantum advantage requires:

- computational scaling beyond classical methods
- or access to structures difficult to simulate classically

Examples:

- large Hilbert space exploration
- entanglement-driven correlations
- efficient operator inversion

Limitations of NISQ Devices

Current devices:

- noisy
- shallow circuits
- limited qubit connectivity

Implications:

- restricts circuit depth
- affects trainability (barren plateaus)

Noise affects:

$$\langle O \rangle \rightarrow \langle O \rangle + \epsilon$$

- gradient estimation becomes unstable
- optimization landscapes distorted

Most realistic near-term approach:

- classical preprocessing
- quantum feature map / variational circuit
- classical optimization loop

Hybrid learning loop

Expressivity vs Trainability

Tradeoff:

High expressivity $\leftrightarrow$ hard optimization

- deep circuits $\rightarrow$ expressive but unstable
- shallow circuits $\rightarrow$ stable but limited

Quantum Advantage in Kernels

Quantum kernels may outperform classical if:

- feature map is hard to simulate classically
- kernel captures non-classical correlations

Open question:

When is the kernel truly quantum?

Recall:

$$u = \mathcal{L}^{-1}f$$

Future direction:

- learn operators directly
- combine quantum circuits with neural operators

QML methods resemble:

- variational wavefunctions (VQE, QNN)
- Green's functions (HHL)
- correlation functions (QSVM kernels)

Machine learning $\leftrightarrow$ many-body theory

Quantum circuits may be interpreted as:

discretized quantum field theories

- parameters $\rightarrow$ coupling constants
- training $\rightarrow$ renormalization-like flow

Key bottlenecks:

- state preparation
- measurement overhead
- noise accumulation
- classical optimization cost

- fault-tolerant quantum computing
- better ansatz design (adaptive methods)
- quantum-classical co-design
- integration with AI architectures

Quantum Machine Learning as Operator Science

- solving, approximating, and learning operators
- unifying physics and computation

The future lies at the interface of quantum physics, machine learning, and o

- understanding structure is key
- not just algorithms, but principles